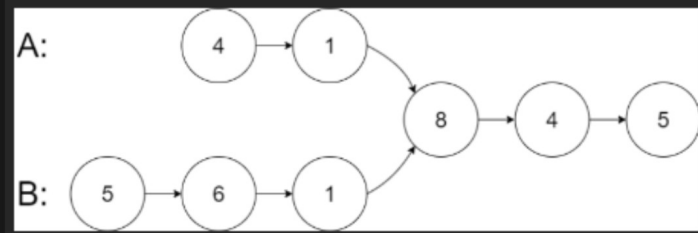


Day-20 Solve Examples on LL

1) Pbm No:- 206 Reverse the linked list

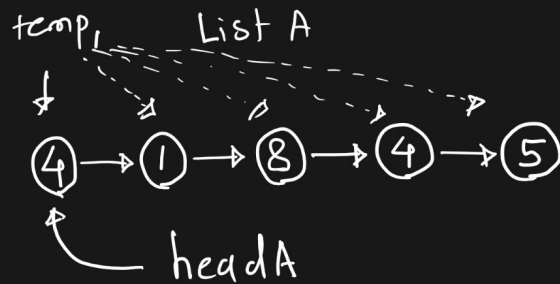
2) Pbm No:- 160

Example 1:

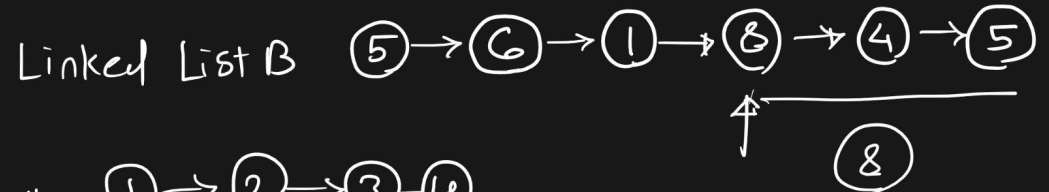
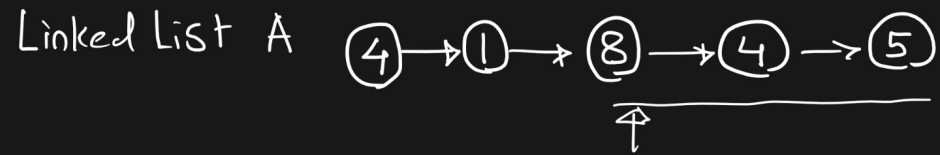
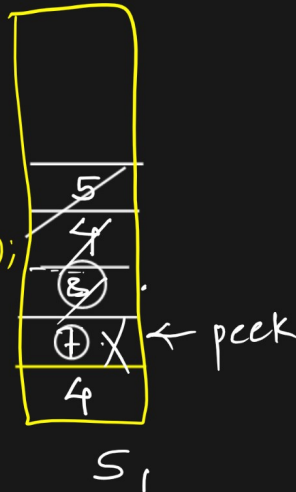


data next

4	add
---	-----



```
ListNode temp = head A;
Stack s1 = new Stack<Integer>();
while (temp != null)
{
    s1.push(temp.data);
    temp = temp.next;
}
```



e.g. 1 -> 2 -> 3 -> 4
 5 -> 6 -> 7 -> 8
 null => no intersection

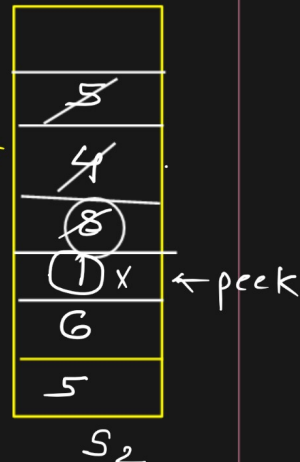
List B



head B `ListNode temp2 = head B;`

`Stack s2 <int> = new Stack<int>();`

```
while (temp2 != null)
{
    s2.push(temp2.data);
    temp2 = temp2.next;
}
```



}

int intersection = -1;

}

while (!s1.empty() & !s2.empty())

{

if (s1.peek() == s2.peek())

X { ~~5~~ ~~4~~ ~~8~~ 7 ~~5~~ ~~4~~ ~~8~~ 1

intersection = s1.peek();

~~5~~ ~~4~~ (8)
- - -

s1.pop();

s2.pop();

}

else

{ return;

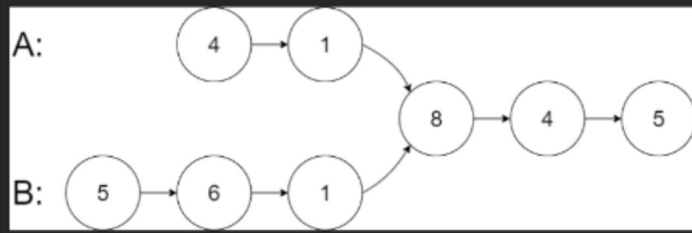
}

return intersection;

}

}

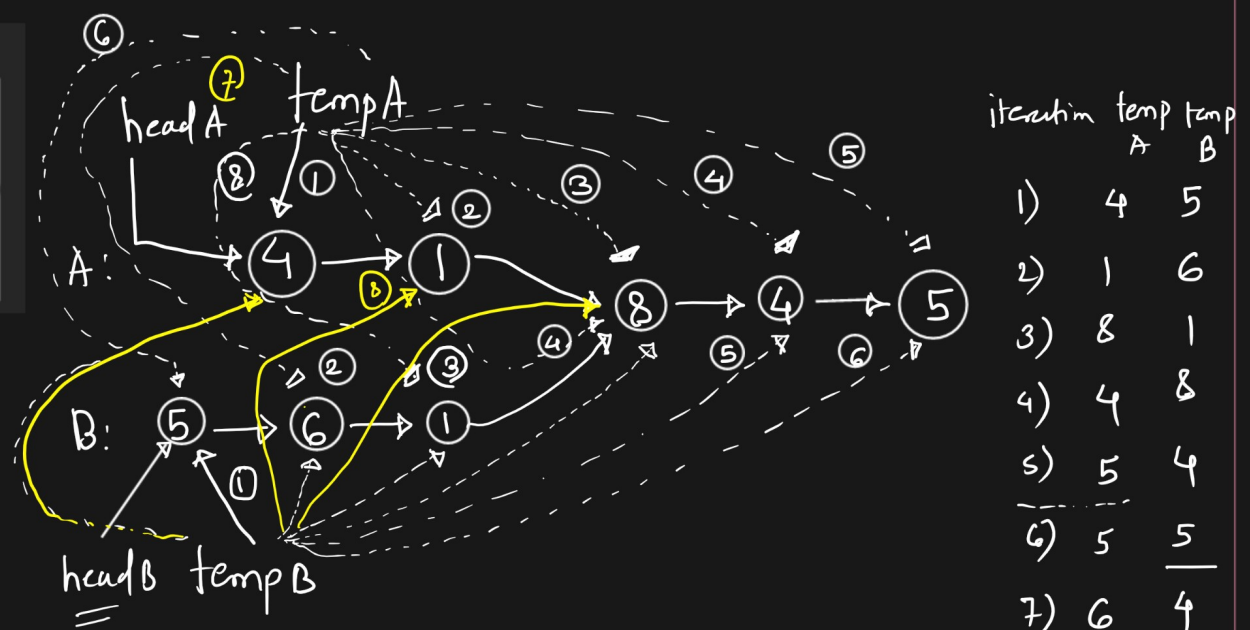
Example 1:



Two Pointer Concepts

{ ListNode tempA = headA

ListNode tempB = headB



iteration	temp A	temp B
1)	4	5
2)	1	6
3)	8	1
4)	4	8
5)	5	4
6)	5	5
7)	6	4
8)	1	1
9)	8	8

while (tempA != tempB)

{ tempA = tempA ? tempA.next : headB;

tempB = tempB ? tempB.next : headA;

}

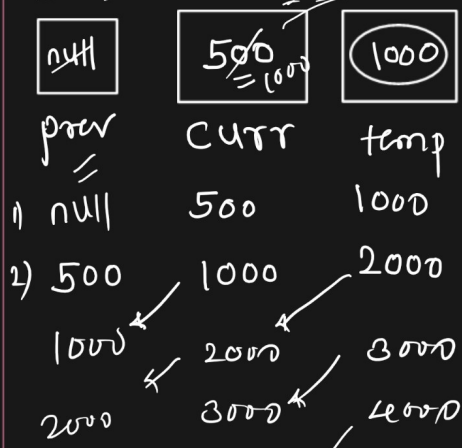
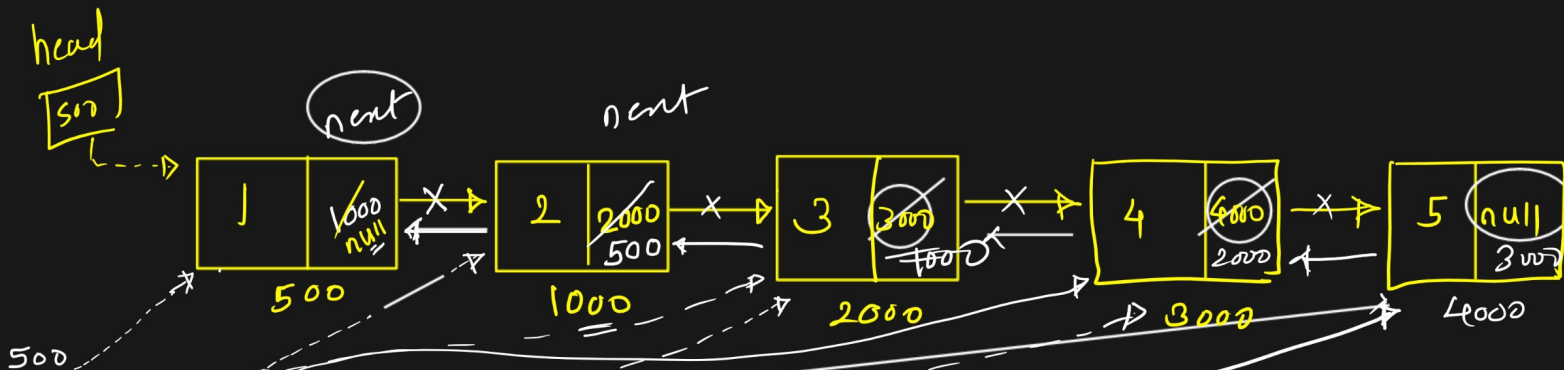
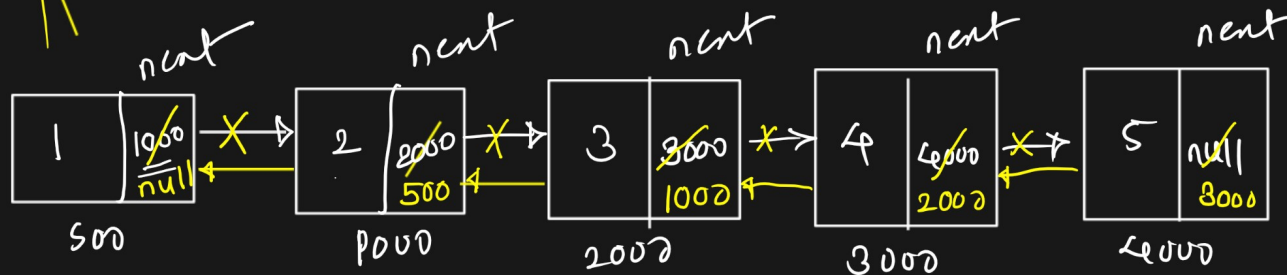
return tempA;

}

Pbm No :- 206 :- Reverse the Linked List

i/p $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

o/p $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$



ListNode prev = null;
ListNode curr = head;
ListNode temp;

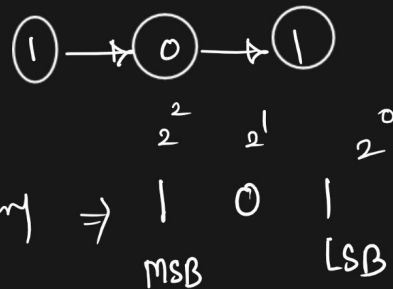
while (curr != null)

```
{
    temp = curr.next;
    curr.next = prev;
    prev = curr;
    curr = temp;
}
```

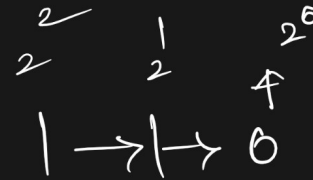
3000 → 4000 → null

return prev;

- ③ Leetcode Ques 876 ⇒ Middle of the linked list } H.W.
 ④ Leetcode Ques 1290 ⇒ Convert binary to decimal }



Decimal ⇒ 4 + 0 + 1
 ⇒ 5



o/p ⇒ 4 + 2
 ⇒ 6